14 JEE maths tricks that skip the long algebra

Every one is from a TutorDA reel. Learn the idea once, use it on a whole family of questions.

01 **How many solutions does $\sin x = x/10$ have?**

$\sin x$ stays in $[-1, 1]$, so only $-10 \le x \le 10$ matters. Sketch both graphs and count crossings.

Answer: 7

02 **How many solutions does $|x| = \cos x$ have?**

$\cos x \le 1$, so $|x| \le 1$. One crossing on the right, and the graphs are symmetric.

Answer: 2 ($x \approx \pm 0.739$)

03 **Which is bigger, 2^{100} or 3^{63}?**

Compare logs, not numbers: $100 \log 2 = 30.10$ and $63 \log 3 = 30.06$.

Answer: 2^{100} (both have 31 digits)

04 **Which is bigger, e^{π} or π^{e}?**

Take ln and divide by πe: compare $\ln x / x$ at e and π. It peaks at $x = e$.

Answer: e^{π} ($23.14 > 22.46$)

05 **How many real roots does $x^3 - 3x + 1 = 0$ have?**

$f' = 0$ at $x = \pm 1$. $f(-1) = 3$ is above the axis, $f(1) = -1$ is below: up, down, up.

Answer: 3 real roots

06 **$\int_0^1 x(1-x)^{99}\,dx = ?$**

Replace x by $1 - x$ ($a + b - x$ property): $\int_0^1 (x^{99} - x^{100})\,dx$.

Answer: $1/100 - 1/101 = 1/10100$

07 **What does $\sin x / x$ approach as $x \to 0$?**

Areas on the unit circle give $\sin x < x < \tan x$, so $\cos x < \sin x / x < 1$.

Answer: 1 (squeezed)

08 **What fraction of the unit square lies under $y = x^2$?**

Rectangle sums close in on $\int_0^1 x^2\,dx$. Not half: the curve hugs the axis.

Answer: 1/3

09 **$1 + 3 + 5 + \ldots + (2n - 1) = ?$**

Each odd number wraps one more L-shape around a square of dots.

Answer: n^2

10 **Two dice: which total comes up most?**

Draw the 6×6 grid of 36 pairs. Totals of 7 fill the whole diagonal.

Answer: 7, with probability $6/36 = 1/6$

11 **Biggest rectangle inside a circle of radius r?**

Corner at angle θ: area $= 4r^2 \sin\theta \cos\theta = 2r^2 \sin 2\theta$, largest at $\theta = 45°$.

Answer: a square, area $2r^2$

12 **Monty Hall: after a goat door opens, switch?**

Your first pick wins 1 time in 3. The other 2/3 moves to the last closed door.

Answer: switch, win 2/3 of the time

13 **$1/(1\cdot2) + 1/(2\cdot3) + \ldots + 1/(99\cdot100) = ?$**

Split each term: $1/(k(k+1)) = 1/k - 1/(k+1)$. Written out, the middle terms cancel in pairs.

Answer: $1 - 1/100 = 99/100$

14 **Which is bigger, 31^{11} or 17^{14}?**

Compare with powers of 2: $31^{11} < 32^{11} = 2^{55}$ and $17^{14} > 16^{14} = 2^{56}$.

Answer: 17^{14} (more than 6 times bigger)